

Lab-10

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System and Network admin

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Submission Guidelines

Submit the following items:

- **Screenshot #1:** Phase 1: Creation, formatting, and mounting of the loopback storage volume.

```
the old version; use 'systemctl daemon-reload' to
server@server:~$ df -h /srv/shared
Filesystem      Size  Used Avail Use% Mounted on
/dev/loop0      974M   24K  907M   1% /srv/shared
server@server:~$
```

- **Screenshot #2:** Phase 3: Backup duration output and the tar.gz archive file size details.

```
/dev/loop0      974M   24K  907M   1% /srv/shared
server@server:~$ sudo mkdir -p /srv/shared/dept
server@server:~$ echo "important report" | sudo tee /srv/shared/dept/report.txt
important report
server@server:~$ echo "student data sample" | sudo tee /srv/shared/dept/student_sample.txt
student data sample
server@server:~$ sudo find /srv/shared -type f -exec sha256sum {} \; | sort | sudo tee /tmp/shared_checksums_before.txt
b7ddf9e5acf630b46e960d998e8c0670bdf2c42de7f2fbd1cc3f7276c5b92f9c  /srv/shared/dept/report.txt
cdcde0a1a4b4ad9b121ee897a725c9f502036e3f960523e245b2fc50f761915  /srv/shared/dept/student_sample.txt
server@server:~$ sudo mkdir -p /backup
server@server:~$ START=$(date +%s)
server@server:~$ sudo tar -czf /backup/shared-backup.tar.gz -C /srv/shared .
server@server:~$ END=$(date +%s)
server@server:~$ echo "Backup duration: $((END-START)) seconds"
Backup duration: 27 seconds
server@server:~$ ls -lh /backup/shared-backup.tar.gz
-rw-r--r-- 1 root root 258 Jun  2 09:12 /backup/shared-backup.tar.gz
server@server:~$
```

- **Screenshot #3:** Phase 5: Restore duration and the empty output of the diff checksum comparison.

```
total 4
-rw-r--r-- 1 root root 20 Jun  2 09:11 student_sample.txt
server@server:~$ START=$(date +%s)
server@server:~$ sudo tar -xzf /backup/shared-backup.tar.gz -C /srv/shared
server@server:~$ END=$(date +%s)
server@server:~$ echo "Restore duration: $((END-START)) seconds"
Restore duration: 15 seconds
server@server:~$ sudo find /srv/shared -type f -exec sha256sum {} \; | sort | sudo tee /tmp/shared_checksums_after.txt
b7ddf9e5acf630b46e960d998e8c0670bdf2c42de7f2fbd1cc3f7276c5b92f9c  /srv/shared/dept/report.txt
cdcde0a1a4b4ad9b121ee897a725c9f502036e3f960523e245b2fc50f761915  /srv/shared/dept/student_sample.txt
server@server:~$ diff /tmp/shared_checksums_before.txt /tmp/shared_checksums_after.txt
server@server:~$
```

- **Reflection Report:** Written answers to the Phase 6 RTO/RPO reflection and the questions below.

Phase 6: RTO / RPO Reflection

- **Measured Restore Time:** The restore duration is whatever you observed in your lab output (e.g., 3–5 seconds for a small dataset). This is the actual time it took to extract the tarball back into /srv/shared.
- **RTO (Recovery Time Objective):** Based on the measured restore time, you can claim an RTO of just a few seconds to minutes for file-level recovery in this lab. In practice, RTO is the maximum acceptable downtime — here, restoring a deleted file was nearly instantaneous.
- **RPO (Recovery Point Objective):** The RPO is the point in time of the last backup. If a failure occurred right before the backup, no data is lost. If a failure occurred hours after the backup, any changes made since the backup would be lost. In this lab, the RPO is essentially “the last backup run,” so the maximum data loss window is the time between backups.
- **System Loss Recovery:** If the entire VM or hypervisor were destroyed, restoring just the tarball wouldn’t be enough. You’d need to:
 - Recreate the VM (install Ubuntu again).
 - Re-attach storage or loopback devices.
 - Restore the backup archive from an external/off-site location.
 - Re-verify integrity with checksums.

Phase 7: Real-World Industry Backup Architectures

- **Did the backup protect against accidental deletion?** Yes. When the file was deleted, restoring from the tarball brought it back exactly as before. This shows backups protect against user mistakes.

- **Would this local backup protect against total VM loss?** No. Since the backup tarball is stored on the same VM disk, if the VM or host disk is lost, both the primary data and the backup are gone.
- **What is the RPO of this backup strategy, and how could it be optimized?** The RPO is the time between backups (e.g., one day if you only back up daily). To optimize, you'd increase backup frequency or use continuous replication (e.g., rsync or BorgBackup) to reduce the data loss window.
- **Why is testing data restoration and integrity verification more critical than just verifying backup completion?** Because a backup that "completed successfully" may still be corrupt or incomplete. Only by restoring and comparing checksums can you prove the backup is usable and the data is intact. Integrity testing ensures confidence in recovery.